

Tanmay Sutar

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EDUCATION

Georgia Institute of Technology

M.S. Computer Science — Machine Learning

2023 – 2025

IIT Guwahati

Bachelors of Technology

2019-2023

EXPERIENCE

Applied AI Engineer

Magic Hour · YC W24 — Oakland, CA

Sep 2025 – Present

- Architected multimodal agentic pipeline (FLUX + Depth Anything V2 + Florence-2 + SAM2 + CUDA parallax); Claude-powered planning pass converts prompt/scene metadata into constrained JSON directives for render control.
- Built custom agent orchestrator with staged-tool execution, skip logic, structured state, and deterministic recovery; exposed via async FastAPI + Next.js control surface.
- Shipped multi-turn creative agent (LangGraph) with visual context caching — precomputed scene descriptions cut vision API calls 90%; semantic reference resolution and failure replay hooks.
- Built LDI Workflow: prompt → cinematic video in ~25s on L4 — MiDaS depth → PyTorch 3D point-cloud splatting → NVENC H.264; Gemini 2.5 Flash story planner drives parallel Flux image gen + Kokoro TTS.

RESEARCH

Annotation Techniques for Judo Combat Phase Classification from Tournament Footage

Georgia Tech France, Metz, France · <https://arxiv.org/pdf/2412.07155>

Jan 2024

- Designed an active learning annotation pipeline in **Label Studio** over 110 hours of live-streamed judo footage, iteratively pre-annotating frames with **YOLOv8** and routing uncertain samples to human reviewers for correction.
- Established frame-sampling and labeling workflows (1 fps extraction, 70/15/15 splits) across 10 concurrent mat views, reducing manual annotation cost while maintaining coverage across all match phases.
- Enabled transfer learning of a combat phase classifier via progressively refined bounding-box labels for players and referees, achieving **F1 scores of 0.66–0.87** across three match-state classes.

Subliminal Trait Transfer via Agentic Pipelines

Bluedot AI Safety Technical Sprint

Jan 2026 – Present

- Investigating LoRA-encoded toxicity transfer from teacher to student model via innocuous synthetic data; extends Le & Hobbhahn (2025) to agentic supply-chain threat settings.
- Pipeline: AutoGen toxic data gen → filtering → Llama 3.1 8B + LoRA fine-tuning → eval across toxicity probes, refusal rates, and divergence token analysis.

Steering Deceptive Behavior in LLMs

Independent Research, GTAISI

Jan 2026

- Traced deceptive intent to Layer 18 residual stream via activation probing; non-linear safety gates resist linear steering, suggesting compositional robustness beyond vector arithmetic.
- Discovered 'Stealth Paradox': deceptive prompts trigger fewer static analysis alerts than honest ones — revealing alignment tax in code safety tooling.

Benchmarking LLMs for Multi-Turn Online Harassment Attacks

Social Wellbeing Lab, Georgia Tech · [arXiv:2510.14207](https://arxiv.org/abs/2510.14207)

Jan 2025

- Designed multi-agent harassment benchmark with game-theoretic simulation; jailbreak fine-tuning drove ASR to 95.8–96.9% (vs. 57–64% baseline) with refusal collapsing to ~1% across LLaMA-3.1-8B and Gemini-2.0-Flash.
- Built GPT-4o-as-judge evaluation framework over 8-category harassment taxonomy; turn-level analysis revealed distinct escalation trajectories and Dark Triad behavioral profiles in attacked agents.

SKILLS

Agentic & Infra

LangGraph, LangChain, AutoGen, Claude API, FastAPI, Next.js · vLLM, DeepSpeed, Docker, SLURM, Multi-GPU (H200/A100/L4)

ML & Multimodal

SFT, DPO, RLHF, LoRA/QLoRA, Transformers, PEFT, Unsloth · FLUX, SAM2, Florence-2, MiDaS, PyTorch, Gemini API

Safety & Tools

LLM-as-Judge, HarmBench, Red-Teaming, Activation Probing, Steering Vectors · Python, TypeScript, SQL, Bash, W&B, Git